

10.4 Tangent Planes & Linearization

$$y = mx + b$$

$$y - y_0 = \frac{dy}{dx}(x_0)(x - x_0)$$

$$z - z_0 = \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0)$$

$$\underbrace{L(x, y)}_z = \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(y - y_0) + \underbrace{f(x_0, y_0)}_{z_0}$$

10.4.1

$$1) f(x, y) = 2x^3 + y^2; (1, 2, 6)$$

$$f(1, 2) = 2(1)^3 + (2)^2 = 2 + 4 = 6$$

$$z - z_0 = \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0)$$

$$\frac{\partial f}{\partial x} = 6x^2 \quad \frac{\partial f}{\partial x}(1, 2) = 6(1)^2 = 6$$

$$\frac{\partial f}{\partial y} = 2y \quad \frac{\partial f}{\partial y}(1, 2) = 2(2) = 4$$

$$z - 6 = 6(x - 1) + 4(y - 2)$$

$$10. f(x, y) = \ln(x^2 + y^2) \quad (1, 1, \ln 2)$$

$$\frac{\partial f}{\partial x} = \frac{1}{x^2 + y^2} (2x) \quad \frac{\partial f}{\partial x}(1, 1) = \frac{1}{(1)^2 + (1)^2} (2 \cdot 1) = 1$$

$$\frac{\partial f}{\partial y} = \frac{1}{x^2 + y^2} (2y) \quad \frac{\partial f}{\partial y}(1, 1) = \frac{1}{(1)^2 + (1)^2} (2 \cdot 1) = 1$$

$$z - \ln 2 = 1(x - 1) + 1(y - 1)$$

$$25) f(x, y) = e^{x+y} \quad (0, 0)$$

$$L(0.1, 0.05)$$

$$\frac{\partial f}{\partial x} = e^{x+y} \quad \frac{\partial f}{\partial x}(0, 0) = 1$$

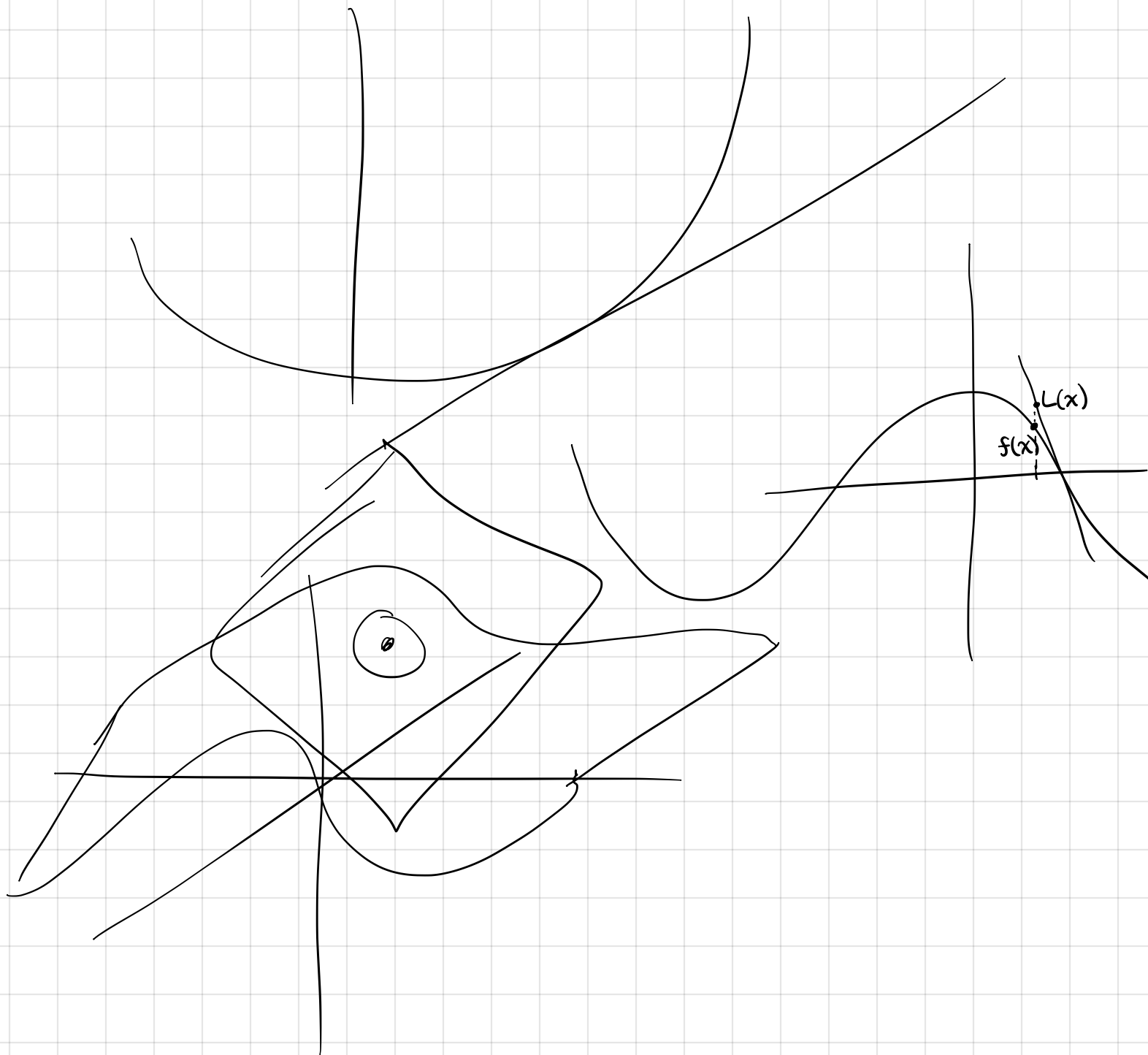
$$\frac{\partial f}{\partial y} = e^{x+y} \quad \frac{\partial f}{\partial y}(0, 0) = 1$$

$$f(0, 0) = e^{0+0} = 1$$

$$\begin{aligned} L(x, y) &= 1(x-0) + 1(y-0) + 1 \\ &= x + y + 1 \end{aligned}$$

$$L(0.1, 0.05) = 0.1 + 0.05 + 1 = 1.15$$

$$f(0.1, 0.05) \approx 1.1618 \quad \leftarrow$$



10.4.2 Jacobi matrices

$\vec{f}(x, y) = \begin{bmatrix} f_1(x, y) \\ f_2(x, y) \end{bmatrix}$

denotes vector

$$J\vec{f}(x, y) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}$$

$x \downarrow$ $y \downarrow$

$\leftarrow f_1$
 $\leftarrow f_2$

2a. $\vec{f}(x, y) = \begin{bmatrix} x + y \\ x^2 - y^2 \end{bmatrix}$

$$f_1(x, y) = x + y$$
$$f_2(x, y) = x^2 - y^2$$

$$J\vec{f}(x, y) = \begin{bmatrix} 1 & 1 \\ 2x & -2y \end{bmatrix}$$

$$\underbrace{\vec{L}(\vec{x})}_{m \times 1 \text{ matrix}} = \underbrace{J \vec{f}(\vec{x}_0)}_{\substack{m \times n \\ m \times n \\ m \times 1}} \underbrace{(\vec{x} - \vec{x}_0)}_{\substack{n \times 1 \\ n \times 1 \\ n \times 1}} + \underbrace{\vec{f}(\vec{x}_0)}_{m \times 1}$$

① can only add matrices if

dimensions match

E.g.

$$\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

n inputs

m outputs

$m \times n$
 $\uparrow \quad \uparrow$
 rows columns

② can only multiply if

$$(m \times n)(n \times q)$$

$\uparrow$
 these dimensions match

$$39. \vec{f}(x, y) = \begin{bmatrix} e^{2x-y} \\ \ln(2x-y) \end{bmatrix} \quad (1, 1)$$

$$J\vec{f}(x, y) = \begin{bmatrix} 2e^{2x-y} & -e^{2x-y} \\ \frac{2}{2x-y} & \frac{-1}{2x-y} \end{bmatrix}$$

$$J\vec{f}(1, 1) = \begin{bmatrix} 2e & -e \\ 2 & -1 \end{bmatrix} \quad \vec{f}(1, 1) = \begin{bmatrix} e \\ 0 \end{bmatrix}$$

$$\vec{L}(x, y) = \begin{bmatrix} 2e & -e \\ 2 & -1 \end{bmatrix} \left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) + \begin{bmatrix} e \\ 0 \end{bmatrix}$$

$$\vec{L}(x, y) = \begin{bmatrix} 2e(x-1) - e(y-1) + e \\ 2(x-1) - \end{bmatrix}$$

$$\vec{L}(x, y) = \begin{bmatrix} ze & -e \\ z & -1 \end{bmatrix} \left(\underbrace{\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\begin{bmatrix} x-1 \\ y-1 \end{bmatrix}} \right) + \begin{bmatrix} e \\ 0 \end{bmatrix}$$

$$\vec{L}(x, y) = \begin{bmatrix} ze(x-1) - e(y-1) + e \\ z(x-1) - (y-1) + 0 \end{bmatrix}$$

10.5.1 Multivariate Chain Rule

$$w = f(x, y)$$

$$x(t)$$

$$y(t)$$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$f(x, y) = x^2 + y^2 \quad x(t) = 3t \quad y(t) = e^t$$

$$\frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = 2y \quad \frac{dx}{dt} = 3 \quad \frac{dy}{dt} = e^t$$

$$2x \cdot 3 + 2y \cdot e^t$$

$$2 \cdot 3t \cdot 3 + 2e^t \cdot e^t$$